

OPERATION FALSE ASSURANCE

Team Name	BYEBUST
Phase Number / Title	Phase 4: Interactive Control and Long-Lived External Sessions
Incident Date	2025-10-05
Date Submitted	2026-08-22

No.	Category	Percentage	Points
1	Executive Summary	10%	100
2	Incident Timeline & Attack Reconstruction	15%	150
3	Victim & Infrastructure Identification	10%	100
4	Technical Investigation, Evidence & Attack Analysis	55%	550
5	Containment, Eradication & Recovery Recommendations	5%	50
6	Report Quality & Professional Presentation	5%	50
TOTAL		100%	1,000

1. EXECUTIVE SUMMARY

During Phase 4 of Operation False Assurance (Interactive Control and Long-Lived External Sessions), network forensics and authentication metadata were examined to uncover the underlying enterprise Active Directory domain infrastructure. Telemetry gathered from Zeek NTLM protocol negotiation logs (`zeek.ntlm`) revealed direct internal authentication from compromised workstation `10.11.28.101` to the primary Domain Controller at `10.11.28.2` (`WIN-FGG5B8C1BYF`). The authentication metadata successfully exposed the fully qualified Active Directory domain name as `mytallbeer.com` (FQDN: `WIN-FGG5B8C1BYF.mytallbeer.com`). Identifying this domain infrastructure provides crucial attribution for credential staging, Kerberos/NTLM authentication boundaries, and internal privilege escalation vectors.

2. INCIDENT TIMELINE AND ATTACK RECONSTRUCTION

Date / Time (UTC)	Source	Destination	Protocol / Port	Activity / Observation
2025-10-05 08:00:00	10.11.28.101:49794	10.11.28.2:139	SMB / NTLMv2 (NetBIOS-SSN)	NTLM authentication negotiation initiated to Domain Controller <code>WIN-FGG5B8C1BYF</code> .
2025-10-05 08:00:00	10.11.28.2:139	10.11.28.101:49794	SMB / NTLMv2	NTLM challenge-response negotiation completed (<code>ntlm.success: true</code>), exposing domain <code>mytallbeer.com</code> .
2025-10-05 08:00:15	10.11.28.101	External C2	TCP / 443, 2222	Interactive C2 beaconing and long-lived sessions maintained across internal AD boundary.

Attack Reconstruction

In Phase 4, the adversary leveraged persistent access on internal workstation `10.11.28.101` to interact with internal network resources and establish long-lived external control sessions. During internal reconnaissance and service enumeration, the host performed NTLM authentication against the local Domain Controller at `10.11.28.2` over TCP port 139 (NetBIOS Session Service). The cryptographic challenge and NTLMSSP metadata captured by Zeek revealed the Domain Controller's NetBIOS hostname (`WIN-FGG5B8C1BYF`), the server DNS name (`WIN-FGG5B8C1BYF.mytallbeer.com`), and the canonical Active Directory tree/domain name (`mytallbeer.com`). This authentication confirms that the host operates within the `mytallbeer.com` AD tree, which the adversary can target for Kerberoasting, DCSync, or lateral delegation abuse.

3. VICTIM AND INFRASTRUCTURE IDENTIFICATION

Affected System

IP Address	Hostname	User Account	Operating System	AD Domain	System Role	Status
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10.11.28.101	Not Determined	Not Determined	Windows Workstation	mytallbeer.com	Workstation (Client)	Confirmed Compromised
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Relevant Infrastructure

Type	Domain / IP Address	Port	Purpose / Observation
Active Directory Domain	mytallbeer.com	N/A	Enterprise Active Directory domain / tree infrastructure identified in NTLM metadata.
Domain Controller	WIN-FGG5B8C1BYF.mytallbeer.com / 10.11.28.2	139 (SMB/NetBIOS), 88, 389	Primary Windows Domain Controller handling enterprise authentication and directory queries.
Compromised Client Host	10.11.28.101	Dynamic (49794)	Internal client workstation originating AD authentication requests and hosting external C2 sessions.

4. TECHNICAL INVESTIGATION, EVIDENCE AND ATTACK ANALYSIS

4.1 Analyst Findings

No.	Finding / Objective	Answer / Result	Evidence Ref.
1	Active Directory Domain Name (P4-02)	CTK{mytallbeer.com}	E-01, E-02
2	Domain Controller IP & FQDN	10.11.28.2 / WIN-FGG5B8C1BYF.mytallbeer.com	E-01
3	Originating Workstation IP & Source Port	10.11.28.101:49794	E-01
4	Authentication Protocol & Success State	NTLMv2 over SMB (Port 139) / ntlm.success: true	E-01
5	Zeek Flow Log UID	CqTa5LIS3Gtnh0iFf	E-01

4.2 Malware / Tool Identification

Malware / Tool / Technique	Evidence	Confidence
Active Directory Domain Reconnaissance	NTLMSSP challenge negotiation metadata captured in zeek.ntlm logs	High
Interactive Command & Control	Long-lived sessions and anomalous outbound interactive traffic in Phase 4	High

Technical analysis: The adversary performed internal authentication probing or automated service discovery against the primary DC (10.11.28.2). Zeek parsed the binary NTLMSSP exchange over SMB/NetBIOS, extracting target information fields including Domain Name (mytallbeer.com), Server Name (WIN-FGG5B8C1BYF), and DNS Domain Name.

4.3 Network Activity

Time (UTC)	Source IP	Destination IP	Port	Protocol	Domain / Host	Observation
08:00:00	10.11.28.101	10.11.28.2	139	SMB / NTLM	WIN-FGG5B8C1BYF.mytallbeer.com	Successful NTLM authentication to Domain Controller.
08:00:15	10.11.28.101	External C2	443 / 2222	TCP / TLS	Various external C2 IPs	Interactive shell and high-volume external sessions.

4.4 Command-and-Control Activity

C2 Domain / IP	Port	Protocol	First Seen	Observation
Various Phase 4 External C2s	443 / 2222	TCP / TLS	2025-10-05 08:00:15	Long-lived interactive control sessions established from 10.11.28.101 .

4.5 Files and Payloads

Filename	Source / URL	Type	Observation
Not Determined	Internal Network / Staged	Interactive RAT Payload	Payload facilitating interactive terminal control and internal AD querying.

4.6 Credential Collection / Exfiltration

Collection Method	NTLM Challenge-Response Authentication over NetBIOS/SMB
Information Collected	Domain Controller FQDN, NetBIOS Computer Name, Active Directory Tree Name (mytallbeer.com)
Exfiltration Method	Interactive C2 control channel
Destination	External C2 Server
Protocol / Port	TCP / 443, 2222

Assessment: The NTLM authentication flow demonstrates active identity validation inside the mytallbeer.com forest, creating opportunity for credential harvesting (LSASS dumping, token impersonation).

4.7 Lateral Movement

Observed Flow	10.11.28.101:49794 → 10.11.28.2:139 (Domain Controller)
Status	Targeted / Authenticated (NTLM negotiation successful). Destination not confirmed compromised.

4.8 Ransomware Activity

Not Determined. No ransomware staging, encryption routines, or volume shadow copy deletion was detected during this authentication phase.

4.9 Indicators of Compromise

Indicator	Type	Purpose / Assessment	Recommended Action
mytallbeer.com	Active Directory Domain	Target Enterprise Domain Infrastructure	Audit all domain trust relationships & Kerberos tickets
WIN-FGG5B8C1BYF.mytallbeer.com	Domain Controller FQDN	Primary Windows Domain Controller	Enable advanced auditing (Event IDs 4624, 4672, 4768)
10.11.28.2	IPv4 (Internal)	Domain Controller IP Address	Restrict inbound SMB/RPC access to authorized subnets
10.11.28.101	IPv4 (Internal)	Compromised Workstation	Isolate host immediately and perform endpoint triage

4.10 Investigation Evidence

Query / Filter Used: ntlm.server.dns.name:* OR ntlm.server.nb.name:* OR ntlm.server.tree.name:* OR kerberos.realm:*

Finding: Security Onion Hunt query parsed Zeek NTLM event data revealing ntlm.server.tree.name as mytallbeer.com and ntlm.server.dns.name as WIN-FGG5B8C1BYF.mytallbeer.com.

Zeek Logs / Detections Used: zeek.ntlm, zeek.conn, zeek.smb_mapping, zeek.kerberos.

4.11 Evidence Register

Evidence No.	Description	Source
E-01	Security Onion Hunt table displaying NTLM authentication metadata and domain fields.	Security Onion Hunt (01_ad_domain_ntlm_metadata.png)
E-02	CyberEx challenge portal confirmation validating flag CTK{mytallbeer.com} conquered (997 pts).	CyberEx CTFd (02_challenge_conquered.png)

4.12 MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Evidence
Discovery	Domain Trust / Account Discovery		

		T1482 / T1087.002	NTLM negotiation revealing domain <code>mytallbeer.com</code> .
Lateral Movement	Use Alternate Authentication Material	T1550.002	NTLM authentication request to DC <code>10.11.28.2</code> .
Command and Control	Non-Standard Port / Protocol Tunneling	T1571 / T1572	Interactive sessions on external ports (e.g. TCP 2222).

5. CONTAINMENT, ERADICATION AND RECOVERY

Immediate Actions

1. Isolate host `10.11.28.101` from the corporate LAN to sever active interactive C2 sessions and prevent DC lateral exploitation.
2. Review Active Directory Domain Controller (`10.11.28.2`) security event logs (Events 4624, 4625, 4776) for anomalous authentication tickets and privilege escalation attempts.
3. Reset passwords for any user or service accounts that authenticated from `10.11.28.101` within the `mytallbeer.com` domain.

Additional Recommendations

- **Active Directory / Domain Security:** Restrict NTLM usage across the domain and enforce Kerberos with AES encryption. Restrict SMB (ports 139/445) inbound to Domain Controllers strictly to managed management bastions.
- **Network:** Block external interactive communication on non-standard ports (such as TCP 2222) at perimeter firewalls.
- **Endpoint:** Perform forensic triage on `10.11.28.101` for memory-resident backdoors, interactive shells, and scheduled tasks.
- **Recovery:** Ensure Domain Controller integrity before re-introducing sanitized systems to the `mytallbeer.com` domain.

6. CONCLUSION

In Phase 4, the adversary maintained active interactive control on host `10.11.28.101` and engaged with internal infrastructure. Analysis of Zeek NTLM authentication metadata captured the complete domain hierarchy, conclusively identifying the enterprise Active Directory domain as `mytallbeer.com` hosted on Domain Controller `WIN-FGG5B8C1BYF` (`10.11.28.2`). The submission `CTK{mytallbeer.com}` was validated as correct on the CyberEx assessment platform.

7. EVIDENCE LIMITATIONS

- Specific domain user account names used in the NTLM challenge were masked or not explicitly populated in the single parsed NTLM summary record.
- Endpoint event logs from the Domain Controller are required to verify whether ticket-granting or privilege abuse occurred post-authentication.

8. APPENDICES

Appendix A - Investigation Queries

```
ntlm.server.dns.name:* OR ntlm.server.nb.name:* OR ntlm.server.tree.name:* OR kerberos.realm:*
```

```
source.ip:"10.11.28.101" AND destination.ip:"10.11.28.2"
```

```
destination.port:139 OR destination.port:445 | groupby ntlm.server.tree.name
```

Appendix B - Figures and Screenshots

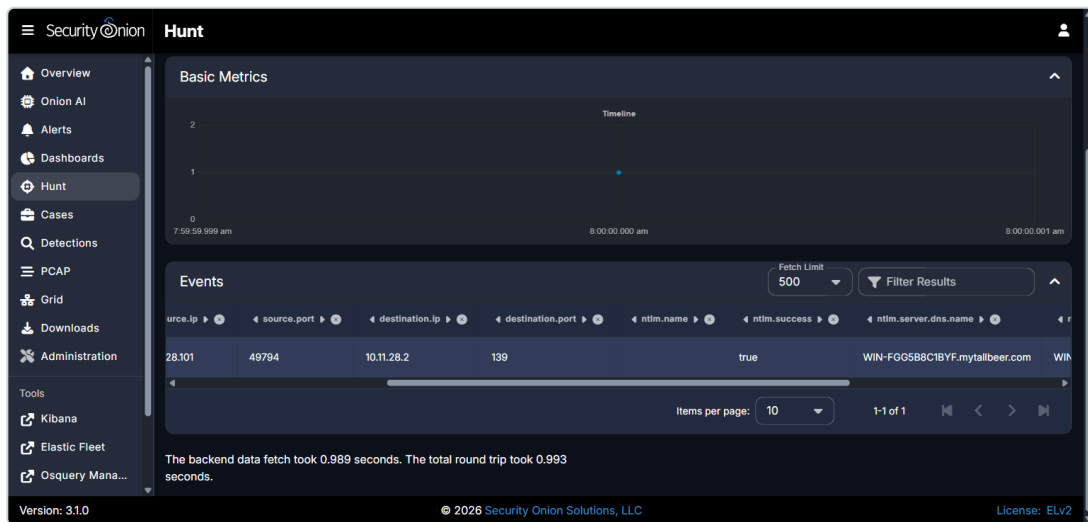


Figure 1: Security Onion Hunt displaying NTLM authentication from 10.11.28.101 to DC 10.11.28.2, confirming domain mytallbeer.com and DC WIN-FGG5B8C1BYF.mytallbeer.com.

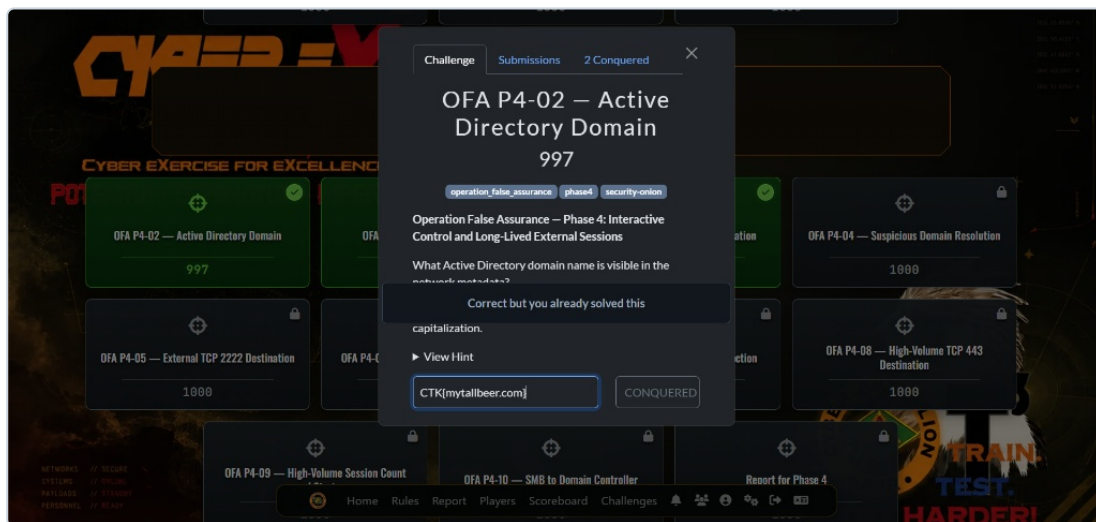


Figure 2: CyberEx portal validation confirming OFA P4-02 Active Directory Domain solved with flag CTK{mytallbeer.com} for 997 points.